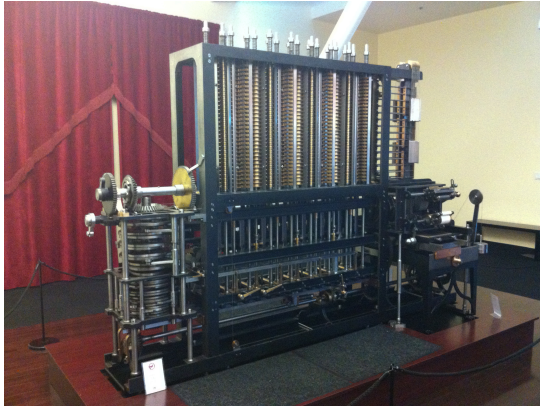




A brief history of computer programming

What counts as a computer?

automatic

programmable

electronic

general-purpose

can play *GTA*

Wrong question: which computer was first?

Right question: first in what sense?

A computer before computers



**Carl Friedrich
Gauss**

1777–1855

Portrait: Gottlieb Biermann, after C. A. Jensen

public domain / Wikimedia Commons

Rechenkünstler

a human calculator

automatic

yes

programmable

yes

electronic

no

general-purpose

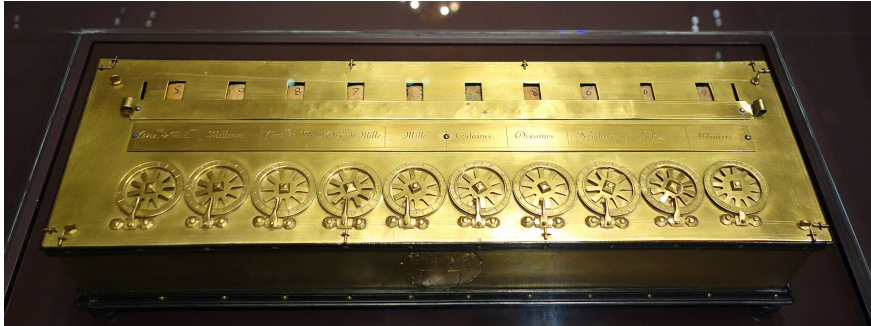
yes

can play GTA?

...

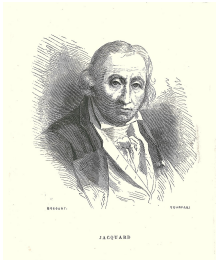
Before machines, a **computer** was a person who computed.

Pascal's calculator, c. 1642

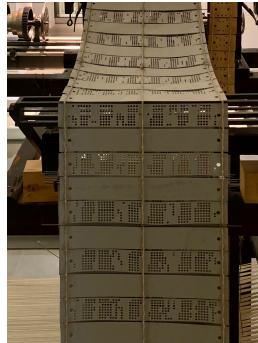


Turn a dial; gears carry the tens.
Mechanical arithmetic, not programmable.

Jacquard loom, early 1800s



**Joseph Marie
Jacquard**
1752–1834



Portrait: 1878 engraving / public domain. Cards: Robert-brook / CC0. Wikimedia Commons

Charles Babbage



Charles Babbage

1791–1871

Reform of math and no music

- ▶ Pushed British mathematics toward continental methods.
- ▶ Designed machines to eliminate errors in printed tables.
- ▶ Campaigned against London's street musicians.

Portrait: Antoine Claudet / public domain / Wikimedia Commons

Babbage's engines

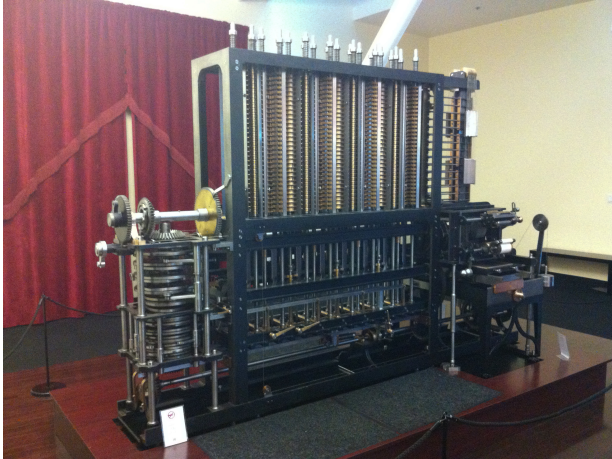


Photo: Victor Grigas / CC BY-SA 3.0 / Wikimedia Commons

1822

Difference Engine

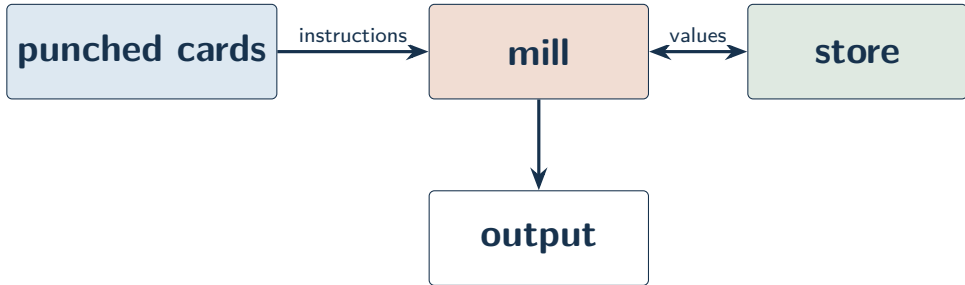
automatic tables

1834 onward

Analytical Engine

a general-purpose design

The Analytical Engine



Lord Byron



**George Gordon
Byron**
1788–1824

What's he doing here?

*My daughter! with thy name
this song begun—*

*My daughter! with thy name
this much shall end—*

Childe Harold's Pilgrimage, III, 115

Annabella Byron



**Anne Isabella
Milbanke**

1792–1860

- ▶ Studied mathematics herself.
- ▶ Byron called her the *Princess of Parallelograms*.
- ▶ Then she married him (probably a mistake)
- ▶ Made mathematics central to their daughter's education.

Portrait: George Hayter / public domain / Wikimedia Commons

Ada Lovelace



Ada Lovelace

1815–1852

- ▶ Wrote the first-ever program (for Analytical Engine).
- ▶ “The Analytical Engine weaves algebraic patterns just as the Jacquard loom weaves flowers and leaves”
- ▶ “The Engine... has no pretensions whatever to originate anything”

Turing's universal machine



Alan Turing

1912–1954

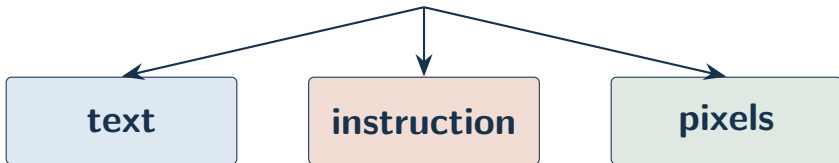
One fixed machine reads
the description of
another.

$$U([M], w) = M(w)$$

Portrait: Elliott & Fry / public domain / Wikimedia Commons

The same bits, different meanings

01000001 01100100 01100001



Meaning comes from the **interpreter**.

Enigma and the Bombe

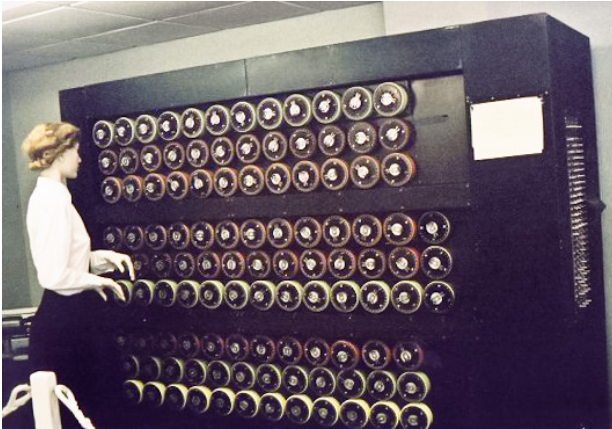


Photo: Sarah Hartwell / CC BY 2.5 / Wikimedia Commons

Bombe

- ▶ searches possible Enigma settings
- ▶ electromechanical
- ▶ specialised

Turing's ACE: design to machine



1945

Turing proposes the Automatic Computing Engine.

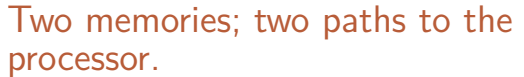
1950

A smaller **Pilot ACE** runs at the National Physical Laboratory.

The full-scale ACE followed in 1958.



Mark I: User B4 / CC BY-SA 2.5. Diagram: Nessa los / CC BY-SA 3.0. Wikimedia Commons

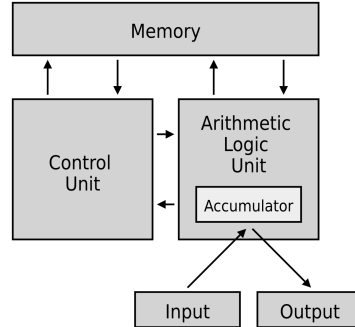


The stored program



John von Neumann

1903–1957



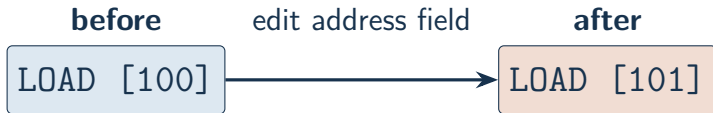
Instructions and data share one memory.

EDVAC draft, 1945.

When code changes itself

Suppose an early machine has **no indexed addressing**.

To print a list, the program must change the address inside an instruction.



memory[100] = first item

memory[101] = second item

The next fetch reads the **new instruction**.

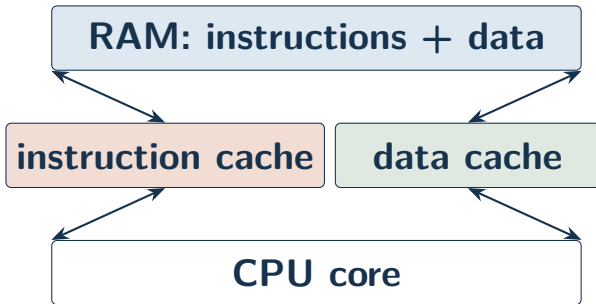
Programs are data, too

A program can inspect, create, or change another program.

- ▶ **Compiler:** writes executable instructions.
- ▶ **Debugger:** reads and sometimes patches them.
- ▶ **Operating system:** loads programs into memory.

Shared-memory stored-program machines make this natural.

Your laptop: a hybrid



Von Neumann in RAM; **Harvard-ish** near the CPU.

Example: Intel Core Ultra 200H/200U processor datasheet, cache hierarchy

“

**What wondrous new
machines
have late been spinning!**

Lord Byron, *Don Juan*, Canto I (1819)